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Effect of 25-OH-cholecalciferol supplementation on acute phase reactants, cytokines, and colostrum quality of Holstein cows during the transition period and plasma metabolites of newborn calves

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ABSTRACT

This study investigated the effects of 25-hydroxycholecalciferol (HyD) supplementation on (1) plasma acute phase reactants, cytokines, and Ig concentrations in colostrum in multiparous Holstein cows from 3 wk prepartum to 21 DIM and (2) plasma concentrations of fat-soluble vitamins, Ig concentrations, minerals, and metabolites of newborn calves. Forty-two cows were assigned to receive either cholecalciferol (D_3) or HyD. From 21 d before the expected calving until parturition, the D_3 group received 0.475 mg/d (0.035 mg/kg DM) of D_3, and the HyD group received 1.45 mg/d (0.108 mg/kg DM) of HyD. From parturition until 21 DIM, the D_3 group received 0.625 mg/d (0.024 mg/kg DM) of D_3, and the HyD group received 0.625 mg/d (0.024 mg/kg DM) of D_3 plus 1.0 mg/d of HyD (0.056 mg/kg DM, equal to 0.080 mg/kg of DM in total). Tail blood samples were collected from cows on d −7 (6 ± 2 d), −3 (2 ± 1 d), 0 (13 ± 8 h postcalving), 1 (36 ± 11 h postcalving), 10 (9 ± 2 d), and 20 (16 ± 2 d) relative to calving for plasma analysis of fat-soluble vitamins (A and E), acute phase proteins (haptoglobin [Hap] and serum amyloid A [SAA]), and cytokines (tumor necrosis-α [TNF-α], IFN-γ, and IL-10). Additionally, a blood sample was taken from a jugular vein of the calves 48 ± 3 (mean ± SD) h after birth. For plasma TNF-α concentration in cows, there was an interaction between treatment and time with lower prepartum concentrations in HyD compared with D_3; however, the difference was reduced at calving and reached the same level as D_3 group at 20 DIM. No treatment effect was observed for plasma Hap, SAA, IFN-γ, or IL-10. Plasma 25-OH-vitamin D_3 (25-OH-D_3) concentration of newborn calves in HyD group was higher than in calves from D_3 group (13.9 and 8.9 pg/mL, respectively), whereas plasma 25-OH-D_2, vitamin A, and vitamin E concentrations were unaffected by the treatment. There was a linear relationship between plasma 25-OH-D_3 in calves and in their dams. Colostrum IgA, IgM, and IgG concentrations were not affected by treatment. In line with the colostrum results, plasma IgA, IgM, and IgG of calves were not different between D_3 and HyD groups. There was no significant difference in plasma Ca and inorganic P concentrations between D_3 and HyD groups in newborn calves. In conclusion, HyD supplementation had limited impact on acute phase reactants and cytokines in transition dairy cows. Dietary supplementation of HyD during the close-up period increased plasma 25-OH-D_3 concentrations in the newborn calves without affecting plasma Ca and inorganic P. Additionally, the increased plasma 25-OH-D_3 concentrations did not affect Ig concentrations in either colostrum or calf plasma.

Key words: acute inflammatory markers, calcidiol, calcitriol, colostrum, immunoglobulin

INTRODUCTION

Cholecalciferol (vitamin D_3) is a fat-soluble vitamin commonly supplemented for dairy cattle diets, which plays critical roles in mineral metabolism and homeostasis, especially in relation to Ca and P (Nelson et al., 2010a). Plasma concentrations of vitamin D metabolites are responsive to supplementary levels of D_3 in the diet. Dietary supplementing with 25-hydroxycholecalciferol (**HyD**) is more effective at raising serum $1,25(OH)_2D_3$ concentrations than using cholecalciferol (Rodney et al., 2018; Poindexter et al., 2020). Rodney et al. (2018) observed that even when cholecalciferol was provided at nearly 4 times the NRC (2001) guidelines, plasma 25-OH-vitamin D_3 (**25-OH-D_3**) concentrations remained below 100 ng/mL, suggesting a regulatory mechanism in the liver's conversion process. Boosting vitamin D status, as indicated by plasma 25-OH-D_3 concentrations (Nelson et al., 2016a), is more easily accomplished with HyD supplementation compared to D_3 supplementation

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

(Poindexter et al., 2020, 2023). Several studies have demonstrated that supplementation with HyD during the close-up period increases plasma concentrations of 25-OH-D_3 (Vieira-Neto et al., 2021; Silva et al., 2022). However, discontinuation of HyD supplementation after calving leads to a depletion of plasma 25-OH-D_3 (Guo et al., 2018), with levels falling below 100 ng/mL, a threshold suggested as indicative of a positive influence of 25-OH-D_3 on immune function (Nelson et al., 2010a).

Nelson et al. (2016a) assessed the serum 25-OH-D_3 concentration in newborn calves and beef cattle. The authors reported that the average serum 25-OH-D_3 concentration in calves at birth was 11 ng/mL, although in most calves, serum 25-OH-D_3 concentrations ranged between 3 and 17 ng/mL. This low plasma concentration of 25-OH-D_3 in calves at birth was observed, even though the plasma 25-OH-D_3 concentrations in the dams (54 ng/mL) were above the assumed sufficiency threshold. The low plasma concentration of 25-OH-D_3 in calves at birth is likely due to the substantial placental barrier that restricts the transfer of fat-soluble vitamins. In addition, newborn ruminants have very low plasma lipid levels because only a small amount of fat crosses from maternal to fetal circulation (Leat, 1967; Pazak and Scholz, 1996), which may further limit the transport of 25-OH-D_3 via lipoproteins.

High-yielding dairy cows are unable to maintain normal blood Ca and P concentrations at calving. This low Ca status, known as hypocalcemia, remains a significant issue, impacting both the production and health of these cows. Cows show acute hypocalcemia when blood total calcium levels fall below 4.5 mg/dL with muscle paresis symptoms, whereas concentrations between ~8.0 and 8.6 mg/dL are classified as subclinical hypocalcemia (NASEM, 2021). Low Ca status may have an adverse effect on the immune system at calving time. Izquierdo et al. (2014) reported that ionized Ca is a vital secondary messenger in immune cells like T and B lymphocytes, macrophages, and mast cells, impacting various signaling pathways that control immune responses. Similarly, the Ca-sensing receptor on immune cells regulates intracellular signaling and cytokine production, playing a key crucial role in maintaining immune balance and presenting a potential target for treating immune-related disorders (Liu et al., 2021).

Cholecalciferol metabolites, particularly 1,25$(OH)_2D_3$, play a key role in regulating innate immunity. Immune cells express 1α-hydroxylase, allowing local synthesis of 1,25$(OH)_2D_3$ (Waters et al., 2001). Binding of 1,25$(OH)_2D_3$ to the vitamin D receptor activates monocytes, macrophages, and neutrophils, enhancing chemotaxis, phagocytosis, and bactericidal activity. Vitamin D also modulates Toll-like receptor and Cluster of Differentiation 14 (CD14) expression, promotes conversion of 25-OH-D_3 to its active form, and induces antimicrobial peptides such as cathelicidin, which targets both gram-positive and gram-negative bacteria. Activated human macrophages can produce 1,25$(OH)_2D_3$ at inflammatory sites (Liu et al., 2006), influencing cytokine and acute phase protein gene expression in immune and hepatic cells. In dairy cow peripheral blood monocytes, pathogen-associated molecules stimulate local 1α-hydroxylase expression, and supplementation with 1.4 ng/mL 25-OH-D_3 increases inducible nitric oxide synthase and RANTES expression in monocytes during LPS challenge (Nelson et al., 2010a). In healthy human immune cells following ex vivo stimulation with bacterial ligands, 1,25$(OH)_2D_3$ reduces proinflammatory cytokines (TNF-α, IFN-γ, and IL-1β) and IL-8, while increasing IL-10 in HK19F-stimulated monocytes (Hoe et al., 2016).

According to the direct and the indirect effect of vitamin D_3 via Ca metabolism on the immune response, and considering that nearly all cows experience some degree of inflammation during the transition period, regardless of their overall health status (Humblet et al., 2006; Horst et al., 2021), we hypothesized that HyD supplementation during the transition period (21 d before expected calving until 21 DIM) would improve acute phase reactants and cytokine activity in dairy cows by maintaining plasma 25-OH-D_3 concentrations above the threshold required for optimal immune function. This effect was expected to be reflected in higher Ig concentrations in colostrum and consequently in plasma of colostrum-fed newborn calves. Furthermore, we hypothesized that HyD supplementation, compared with D_3 supplementation during the close-up period, would improve the vitamin D status of newborn calves. Accordingly, the objective of the present study was to investigate the effect of HyD supplementation on (1) acute phase proteins (haptoglobin [**Hap**] and serum amyloid A [**SAA**]) and cytokines (TNF-α, IFN-γ, and IL-10) during the transition period and Ig concentrations in colostrum and (2) plasma vitamin D status, Ig concentrations, minerals, and metabolites in newborn calves.

MATERIALS AND METHODS

The present study was carried out under the animal experimentation permit number 2020-15-0201-00709 granted by the Animal Experiments Inspectorate of the Ministry of Food, Agriculture and Fisheries of Denmark, in accordance with Justice Law No. 474 (November 2014) concerning animal experiments and the care of experimental animals. This legislation aligns with EU standards for the protection of animals used for scientific purposes.

Animals and Experimental Design

Forty-two multiparous Danish Holstein cows (30 second parity and 12 older cows balanced between the 2 treatments) from Aarhus University were used. Parties in older cows were 3 ± 1 (average ± SD). During the final week of the previous lactation, milk yield was 31.9 kg in the HyD group and 29.6 kg in the D_3 group, and BW was 723.8 and 738.1 kg in the HyD and D_3 groups, respectively. During the close-up period, cows were kept in a loose-housing system with straw bedding, and during the lactating period, cows were kept in a loose-housing system with a concrete floor and cubicles. All cows received the same far-off diet, and from 21 d before expected calving date (22 ± 5 d), cows were assigned to cholecalciferol or supplemented with 25-OH-cholecalciferol. During the close-up period, the D_3 group received 0.475 mg/cow per day (0.035 mg/kg diet DM) of D_3 through vitamin and mineral supplements, and the HyD group received 1.45 mg/cow per day of HyD (0.108 mg/kg diet DM) without D_3 in their vitamin and mineral supplements. After calving, the D_3 group received the recommended vitamin D_3 level of 0.625 mg/cow per day (0.024 mg/kg diet DM), while the HyD group received the D_3 diet supplemented with 1.0 mg/cow per day of HyD (0.056 mg/kg diet DM). All cows were fed the same close-up and lactating TMR diets, except for the vitamin D supplementation (Table 1) formulated to meet the requirement according to the Nordic feed evaluation system (Volden, 2011).

Experimental Sampling and Data Collection

All cows had ad libitum access to feed (7%–10% orts), with feed provided twice daily at 0800 and 1600 h. They also had unrestricted access to water. Daily feed intake was monitored throughout the experiment using Insentec feed troughs (Insentec, Marknesse, the Netherlands). Blood sampling d were scheduled for d −7 (6 ± 2 d), −3 (2 ± 1 d), 0 (13 ± 8 h postcalving), 1 (36 ± 11 h postcalving), 10 (9 ± 2 d), and 20 (16 ± 2 d) relative to the calving date. Tail blood samples were collected in 9 mL sodium heparin Vacutainers (VACUETTE; Greiner Bio-One International GmbH; Austria) on these days, placed on crushed ice, and centrifuged at 3,000 × *g* for 15 min at 4°C. Plasma was then harvested.

The close-up length was 21 ± 5 d (mean ± SD). At ~2 d prior to the expected calving date, cows were relocated to individual calving pens. After calving, both cows and calves remained in the same pen for a duration of a minimum of 12 h. Calves were allowed to suckle from their dam. Additionally, cows were milked in the calving pen, and a colostrum sample was taken (2.2 ± 1.9 h after calving; mean ± SD), and calves were immediately fed with the colostrum from their own dam. The birth BW of calves was recorded, and 48 ± 3 h (mean ± SD) after birth, a blood sample was taken from the jugular vein, placed on crushed ice, and centrifuged at 3,000 × *g* for 15 min at 4°C. Then plasma was harvested and stored at −18°C.

Plasma Fat-Soluble Vitamin Analysis

The concentrations of 25-OH-D_3 and 25-OH-vitamin D_2 (**25-OH-D_2**) in plasma were analyzed using the method described by Hymøller and Jensen (2011). In short, plasma proteins were precipitated with ethanol and methanol, and triglycerides underwent saponification with potassium hydroxide. The fat-soluble vitamins were extracted into heptane, and the samples were dried with nitrogen before being redissolved in 85% methanol. Vitamin D separation and quantification were performed via reverse-phase gradient HPLC with UV detection at 265 nm, utilizing 1α-hydroxycholecalciferol (Sigma-Aldrich; St. Louis, MO) as the internal standard. Separation and quantification were carried out by HPLC as described by Hymøller and Jensen (2011).

The concentration of α-tocopherol in plasma samples was analyzed using HPLC following saponification and extraction into heptane (Jensen et al., 1998). A volume of 1.00 mL of the sample was mixed with 2.00 mL of 96% ethanol, 0.50 mL of methanol, 1.00 mL of 20% ascorbic acid, 0.30 mL of a 1:1 (wt/vol) KOH-water solution, and 0.70 mL of water. The mixture was saponified at 80°C for 20 min and then cooled in darkness. Tocopherol was extracted with 5.00 mL heptane, and 100 µL of the combined heptane extract was injected into the HPLC. All solvents were of HPLC grade. A Perkin-Elmer HS-5-Silica column (4.0 × 125 mm) was utilized for tocopherol analysis, with a mobile phase of heptane modified with 2-propanol (3.0 mL/L), degassed with helium, and a flow rate of 3.0 mL/min. The peak areas and retention times were compared against external standards from Merck for identification and quantification. Fluorescence detection was conducted at an excitation wavelength of 290 nm and an emission wavelength of 327 nm.

Plasma concentration of retinol was determined in the same heptane extract and with the same HPLC column as the tocopherols (Jensen et al., 1998). The mobile phase was heptane modified with 2-propanol (60 mL/L), degassed with helium, and a flow rate of 3.0 mL/min. The peak areas and retention times were compared against external standards from Merck for identification and quantification. Fluorescence detection was conducted at an excitation wavelength of 344 nm and an emission wavelength of 472 nm.

Plasma Metabolites

Plasma Hap concentration was measured using a spectrophotometric assay (Tridelta Developments Ltd., Co. Kildare, Ireland), and plasma SAA was analyzed using latex agglutination (VET-SAA "Eiken" Reagent, Eiken Chemical CO. LTD, Tokyo, Japan), using an Advia 1800 clinical chemistry analyzer (Siemens Healthcare A/S, Bal-

Table 1. Total mixed ration and nutrient composition of diets[1]

	Close-up diets[2]		Lactating diets[3]	
Ingredient, g/kg DM unless noted	D_3	HyD	D_3	HyD
Grass silage	—	—	179.7	179.7
Corn silage	741.1	741.1	344.2	344.2
Barley	—	—	153.0	153.0
Corn, finely ground	74.9	74.9	—	—
Rapeseed cake, 10.5% fat	104.8	104.8	130.0	130.0
Rapeseed meal, 4% fat	—	—	84.1	84.1
Corn gluten	59.9	59.9	—	—
Sugar beet pellets	—	—	91.8	91.8
Salt	—	—	1.8	1.8
Sodium bicarbonate	—	—	9.9	9.9
Chalk	1.9	1.9	0.7	0.7
Magnesium chloride hexahydrate	4.8	4.8	—	—
Ca chloride dihydrate	5.2	5.2	—	—
Mineral and vitamin premix	7.5	7.5	3.7	3.7
Vitamin A, E, and D supplements	—	—	1.1	1.1
Mineral analysis[4]				
DM, g/kg	363 ± 15	364 ± 12	395 ± 5	392 ± 11
Ca, g/kg DM	4.6 ± 0.4	4.6 ± 0.3	6.8 ± 1.3	7.3 ± 0.3
P, g/kg DM	2.7 ± 0.2	2.8 ± 0.1	4.85 ± 0.4	4.75 ± 0.2
Mg, g/kg DM	2.2 ± 0.2	2.3 ± 0.2	3.0 ± 0.4	3.1 ± 0.1
Potassium, g/kg DM	8.4 ± 0.8	8.6 ± 0.4	12.5 ± 0.7	12.5 ± 0.7
Na, g/kg DM	2.5 ± 0.2	2.5 ± 0.2	4.45 ± 1.9	4.85 ± 1.1
Total S, mg/kg DM	2.0 ± 0.2	2.0 ± 0.1	2.9 ± 0.2	2.9 ± 0.1
Fe, mg/kg DM	262 ± 29	308 ± 36	425 ± 191	435 ± 35
Mn, mg/kg DM	40.6 ± 3.9	41.4 ± 3.8	59 ± 12.7	62.5 ± 3.5
Zn, mg/kg DM	95.8 ± 17.8	100.1 ± 15.0	53.5 ± 6.4	74 ± 12.7
Cu, mg/kg DM	14.5 ± 2.6	14.2 ± 2.8	6.6 ± 1.5	7.6 ± 0.8
Cl, mg/kg DM	9.6 ± 1.1	9.6 ± 1.2	4.7 ± 0.8	5.0 ± 0.4
Calculated DCAD, mEq/kg DM	−73	−65	+202	+213
Nutrient composition				
Energy, MJ/kg DM	7.31	7.31	6.56	6.56
CP, g/kg DM	149.9	149.9	163	163
Protein balance in the rumen, g/kg DM	−2	−2	12	12
Fatty acids, g/kg DM	27	27	26	26
NDF, g/kg DM	310	310	320	320
Starch, g/kg DM	262	262	193	193
Cholecalciferol (vitamin D_3),[5] mg/kg DM	0.035	—	0.024	0.024
25-OH-D_3[5] mg/kg DM	—	0.108	—	0.056

[1]Reported already in Jensen et al. (2026).

[2]The D_3 contained 0.475 mg/cow per day of cholecalciferol. However, in HyD, mineral and vitamin supplements did not contain any cholecalciferol, and HyD was supplied in a dose of 1.45 mg/cow per day.

[3]The D_3 contained 0.625 mg/cow per day of cholecalciferol. However, in HyD, mineral and vitamin supplements contained 0.625 mg/cow per day of cholecalciferol, and HyD was added in a dose of 1.45 mg/cow per day.

[4]In the close-up and lactating diets, 9 and 2 samples per treatment were analyzed (mean ± SD), respectively.

[5]Both cholecalciferol and 25-hydroxycholecalciferol were mixed in mineral and vitamin premix. Composition of mineral and vitamin premix for close-up diet (per kg DM): NaCl 635 g, heat-treated wheat meal 150 g, sugar beet molasses 30 g, vitamin A 416,700 IU, vitamin D_3 208,300 IU, synthetic α-tocopheryl acetate 50,000 IU, copper (II) sulfate 2.083 mg, manganese oxide 2.083 mg, zinc oxide 8.333 mg, calcium iodate 208 mg, sodium selenate 50 mg, and cobalt (II) carbonate 58 mg. Composition of mineral and vitamin premix for lactating diet (per kg DM): calcium carbonate 369 g, NaCl 301 g, magnesium oxide 257 g, calcium magnesium carbonate 8 g, sugar beet molasses 3 g, calcium magnesium carbonate 5 g, vitamin A 600,000 IU, vitamin D_3 190,000 IU, synthetic α-tocopheryl acetate 4,000 IU, manganese (II) oxide 4,000 mg, copper (II) sulfate 1,500 mg, zinc oxide 4,500 mg, calcium iodate 225 mg, coated cobalt (II) carbonate 25 mg, and sodium selenate 50 mg.

lerup, Denmark). Plasma concentrations of TNF-α (Nordic Biosite EKX-C3MWO4-96, Nordic Biosite, Täby, Sweden), IFN-γ (Nordic Biosite EKX-KZMYY2-96, Täby, Sweden), and IL-10 (Nordic Biosite EKX-IKE9OB-96, Täby, Sweden) were measured using ELISA.

Brix value was measured by a Milwaukee refractometer (MA871 Digital Brix Refractometer, Romania), and colostrum and calf plasma IgG (EKX-YF5VZ6-96, Nordic Biosite, Täby, Sweden), IgA (EKX-JEX80W-96, Nordic Biosite, Täby, Sweden), and IgM concentrations (EKX-NXKW0V-96, Nordic Biosite, Täby, Sweden) were assayed using ELISA. Total plasma Ca was determined by the Arsenazo method (Siemens CA_2, Siemens Healthcare Diagnostics Manufacturing Ltd., Dublin, Ireland),

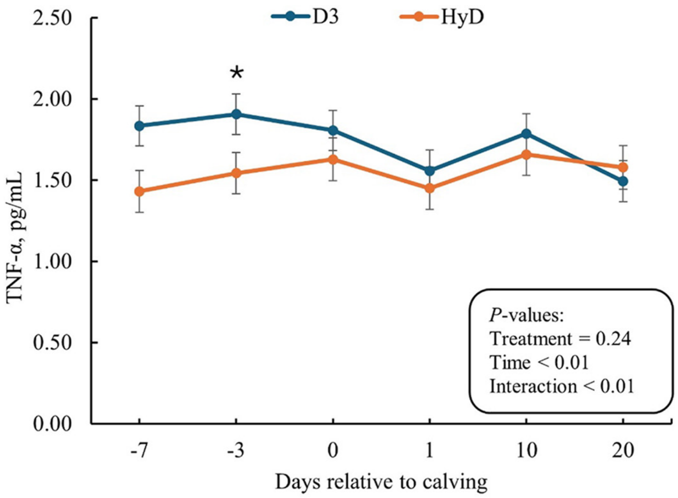

Figure 1. Plasma concentration of TNF-α of cows fed either D_3 (n = 16) or HyD (n = 15) during the transition period. Error bars represent SEM. The CI for treatment was 1.5 to 2.0. Four cows in D_3 and 5 cows in HyD were below detection; therefore, there were 16 and 15 cows in D_3 and HyD, respectively. Data were not normally distributed, and the presented data are transformed data by $\log_{10}$. *Shows significant difference on d −3 relative to calving ($P = 0.05$ for LSM for comparison).

plasma inorganic P concentration was determined by the molybdate method (Siemens IP, Dublin, Ireland), plasma total Mg concentration was determined using the xylidyl blue method (Siemens MG, Dublin, Ireland), plasma total protein concentration was determined using the biuret method (Siemens Total Protein II, Dublin, Ireland), and plasma creatinine concentration was determined using the picric acid method (Siemens CREA, Dublin, Ireland).

Statistical Analysis

Plasma concentrations of fat-soluble vitamins (A and E), acute phase proteins (Hap and SAA), and cytokines (TNF-α, IFN-γ, and IL-10) during the transition period in cows were analyzed using the PROC MIXED procedure in SAS version 9.4 (SAS Institute Inc.), using the maximum likelihood method. The model used was

$$Y_{ijk} = \mu + D_{\mathrm{i}} + T_{\mathrm{j}} + D \times T_{\mathrm{ij}} + c_{\mathrm{k}} + e_{\mathrm{ijk}},$$

where Y is the dependent variable, μ is the overall mean, and the model includes fixed effect of Dth dietary treatment (i; D_3 or HyD), Tth of time (varied in different parameters form 7 d before parturition to 20 DIM), the interaction between dietary treatment and time, and the random effect of kth cow (c_{k}; 20 cows per treatment), and the random error (e_{ijk}). Individual cows were experimental units, and time was also handled as repeated in the model. The approximation for df was specified by DDFM = SATTERTHWAITE option in the stated model. When the interaction between treatment and time or significant effect of treatment was found, LSM were compared within the same days by slice option by using pairwise differences (PDIFF).

Plasma Hap and SAA were not normally distributed and thus were transformed by 1/X and $\log_e$, respectively. Similarly, plasma INF-γ, TNF-α, and IL-10 were transformed by $\log_{10}$.

The calf data (plasma concentration of fat-soluble vitamins, birth BW, plasma concentration of Ig, and other plasma metabolites) and colostrum Ig were analyzed using the PROC MIXED procedure in SAS with maximum likelihood methods, and the model was

$$Y_{ij} = \mu + D_{\mathrm{i}} + c_{\mathrm{j}} + e_{\mathrm{ij}},$$

where Y is the dependent variable, μ is the overall mean, and the model includes the fixed effect of Dth dietary treatment (i; D_3 or HyD), the random effect of jth calf (c_{j}; 20 and 19 calves for D_3 and HyD, respectively), and the random error (e_{ij}). The approximation for df was specified by DDFM = SATTERTHWAITE option in the stated model. Individual calves were experimental units. For analysis of Brix value and colostrum IgA, IgM, and IgG, time for first milking colostrum was included as covariate; however, it was not significant and removed from the analysis.

There were 2 culled cows due to leg and hoof problems postpartum (a cow per treatment), and thus 40 cows were included in the data analysis. Three cows did not produce enough colostrum to feed their calves. Those calves were fed from the colostrum bank, and thus their data were removed from the analysis. Except for Table 1, values presented in tables and figures are LSM with corresponding SEM for normally distributed data or 95% CI for transformed data. Significance was declared at $P \leq 0.05$ and a tendency of $0.05 < P \leq 0.10$.

RESULTS

Plasma Concentrations of Acute Phase Reactants and Cytokines in Cows

Plasma acute phase reactants and cytokines in cows are presented in Table 2. Plasma concentrations of Hap, INF-γ, SAA, and IL-10 in cows were not affected by treatment. Additionally, there was no interaction between treatment and time for plasma Hap, INF-γ, SAA, and IL-10. For plasma concentrations of TNF-α in cows, there was an interaction between treatment and time (Figure 1; $P < 0.01$ for interaction). Plasma TNF-α concentration in cows was greater in the D_3 group compared to the HyD group on d −3 relative to calving ($P = 0.05$ for LSM comparison).

Table 2. Plasma concentrations of acute phase reactants and cytokines of cows fed either D_3 (n = 20) or HyD (n = 20) during the transition period

Item	Treatment		95% CI	*P*-value		
	D_3	HyD		Treatment	Time	Interaction
Hap,[1] g/L						
−7 d before calving	3.7	4.0	2.4–3.0	0.37	<0.01	0.49
−3 d before calving	3.5	4.2				
Calving day	2.5	2.3				
1 d after calving	1.6	1.7				
10 d after calving	2.8	3.1				
20 d after calving	2.6	2.9				
SAA,[1] pg/mL						
−7 d before calving	1.6	1.5	2.0–2.1	0.52	<0.01	0.86
−3 d before calving	1.7	1.5				
Calving day	2.9	2.9				
1 d after calving	3.3	3.3				
10 d after calving	1.9	1.6				
20 d after calving	2.5	2.4				
INF-γ, pg/mL[1]						
−7 d before calving	2.49	2.50	2.5–2.6	0.82	<0.01	0.18
−3 d before calving	2.50	2.48				
Calving day	2.59	2.59				
1 d after calving	2.48	2.54				
10 d after calving	2.45	2.47				
20 d after calving	2.48	2.47				
IL-10,[1] pg/mL						
−7 d before calving	2.11	2.01	1.9–2.3	0.78	0.02	0.63
−3 d before calving	2.05	1.94				
Calving day	2.12	2.16				
1 d after calving	2.15	2.14				
10 d after calving	2.08	2.07				
20 d after calving	2.14	2.13				

[1]Not normally distributed, and the presented data are transformed data. No significant difference was observed between LSM in the same days.

Plasma Concentrations of Vitamin A and E in Cows

The concentrations of plasma vitamin A (Figure 2) and E (Figure 3) in cows were not affected by treatment. For plasma concentration of vitamin E, there was an interaction between treatment and time (P < 0.01). Plasma vitamin E on d −7, −3, 0, and 1 relative to calving was higher in D_3 group than the HyD group; however, on d 20 relative to calving, it was greater in HyD group than the D_3 group.

Concentrations of Fat-Soluble Vitamins, Colostrum Ig, Plasma Ig, and Plasma Metabolites in Newborn Calves

Colostrum yield and characteristics are presented in Table 3. Brix value and colostrum volume were not affected by treatment. Colostrum IgA (P = 0.80), IgM (P = 0.78), and IgG (P = 0.64) levels were not different in D_3 and HyD groups. Birth BW was not affected by treatment (P = 0.92). Plasma concentrations of fat-soluble vitamins, plasma Ig, and plasma metabolites of calves are presented in Table 4. Plasma 25-OH-D_3 concentration of newborn calves from HyD group was 36% greater than that of newborn calves in the D_3 group (P < 0.01). However, plasma 25-OH-D_2 concentration was not affected by treatment. Similarly, plasma vitamin A (P = 0.42) and E (P = 0.64) concentrations of newborn calves were similar between D_3 and HyD groups. Figure 4 shows the relationship between plasma 25-OH-D_3 concentration in newborn calves and their dams, showing that greater plasma 25-OH-D_3 in cows was followed by greater plasma concentrations of 25-OH-D_3 in the newborn calves.

Plasma concentrations of IgA (P = 0.35), IgM (P = 0.96), and IgG (P = 0.84) in newborn calves were similar between D_3 and HyD groups. Similarly, plasma total protein (P = 0.59) and creatinine (P = 0.74) concentrations were not affected by treatment. Additionally, plasma total Ca (P = 0.76), inorganic P (P = 0.66), and Mg (P = 0.50) concentrations in newborn calves were similar between D_3 and HyD groups.

DISCUSSION

Data on DMI, milk yield and composition, plasma 25-OH-D_3 and 1,25$(OH)_2D_3$, arterial blood ionized Ca, plasma mineral, and plasma metabolites have been reported in our previous paper by Jensen et al. (2026).

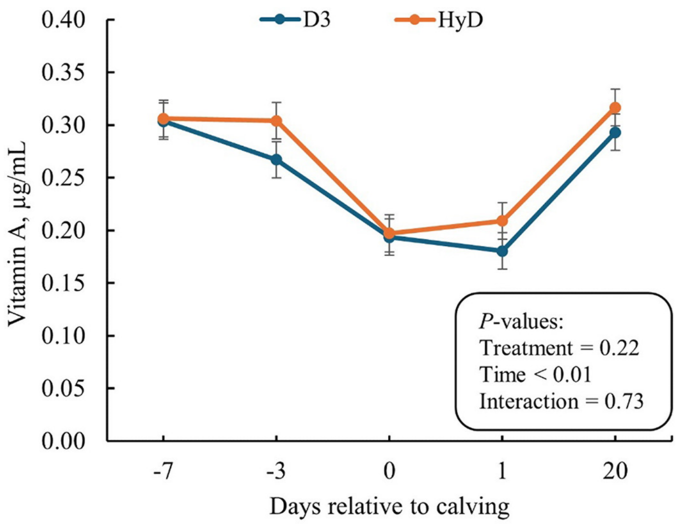

Figure 2. Plasma concentration of vitamin A in cows fed D_3 or HyD. Error bars represent SEM. Treatment SEM was 0.01. No significant difference was observed between LSM in the same days.

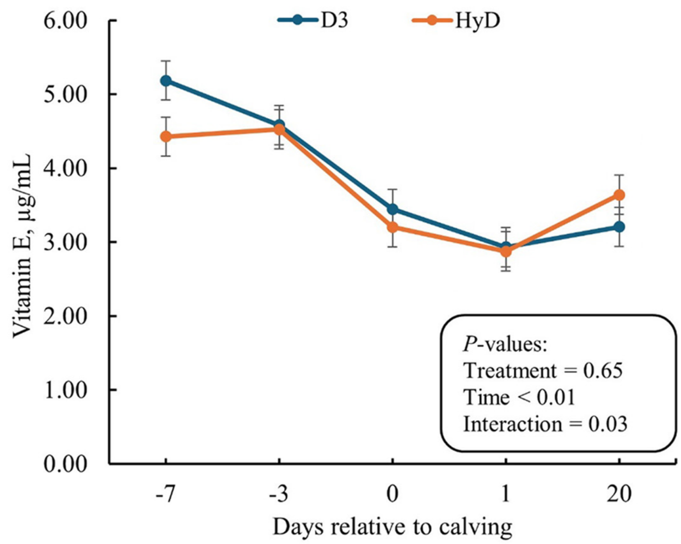

Figure 3. Plasma concentration of vitamin E in cows fed D_3 or HyD. Error bars represent SEM. Treatment SEM was 0.21. No significant difference was observed between LSM in the same days.

Briefly, Jensen et al. (2026) reported that there was more than a 130% increase in plasma concentration of 25-OH-D_3 in HyD group compared to D_3 group on d −7, −3, 0, 1, 10, and 20 relative to calving, without any significant effect on plasma concentration of $1,25(OH)_2D_3$. This increase in plasma concentration of 25-OH-D_3 resulted in a significant increase in arterial blood ionized Ca in HyD group compared to D_3 group on the day of calving without significant effects on plasma concentrations of total Ca, inorganic P, and Mg. Additionally, it is important to note that part of the responses observed in the plasma parameters during lactation between the HyD and D_3 groups could partly be due to the higher supplementation level in HyD group compared to D_3 group.

Plasma Concentrations of Acute Phase Reactants and Cytokines in Cows

Research over the past 10 yr suggests that vitamin D plays a role in enhancing the immune response via enhancing ionized blood Ca level (Vieira-Neto et al., 2017) or by a direct vitamin D effect on the innate immune response (Nelson et al., 2010b). Additionally, according to Humblet et al. (2006) and a review by Horst et al. (2021), nearly all cows experience inflammation during the transition period, regardless of their health condition. Several studies have demonstrated the potential effect of vitamin D on regulation of inflammatory responses (Olliver et al., 2013; Hoe et al., 2016). Hewison (2012), in a review paper, concluded that vitamin D acts to maintain a balance between inflammatory T-helper-1/T-helper-17 cells and immunosuppressive T-helper-2/regulatory T-cells, and sufficient vitamin D within the immune system acts to maintain a tolerogenic immune response by favoring T-helper-2/regulatory versus T-helper-1/T-helper-17 cells. Thus, it was expected that dietary supplementation of 25-OH-D_3 could improve the immune system and inflammatory response of dairy cows during the transition period.

Plasma concentration of 25-OH-D_3 in both pre- and postpartum and blood ionized Ca at the day of calving were greater in the HyD group than the D_3 group (Jensen et al., 2026); however, there was only an interaction between treatment and time for TNF-α. In the present study, plasma acute phase reactants and cytokines in general did not change. In agreement with our findings, Yue et al. (2018) reported no significant effect of sunlight exposure or D_3 supplementation on the number of total leukocytes, lymphocytes, neutrophils, monocytes, and plasma IgA, IgG, and IgM concentrations in healthy late-lactating cows. Additionally, Yue et al. (2018) found no significant effect of sunlight exposure or D_3 supplementation on TNF-α, IL-1, and INF-γ in response to ex vivo stimulation by *Escherichia coli* LPS or *Staphylococcus aureus* peptidoglycan. In contrast to our findings, Nelson et al. (2010b) reported that infected mammary glands with *Streptococcus uberis* led to increased expression of iNOS and RANTES genes in bovine monocytes from both tissue and secreted cells. This upregulation was associated with enhanced induction of the 1α-hydroxylase gene in the presence of 25-OH-D_3 (Nelson et al., 2010a). However, in the present study, no immune response to HyD may reflect the rather healthy cows, which is supported by low disease and metabolic

Table 3. Effect of D_3 (n = 20) or HyD (n = 19) supplementation in the close-up diet of the cows on birth BW of calves and colostrum characteristics

	Treatments			
Item	D_3	HyD	SEM	*P*-value
Birth BW, kg	40.8	41.0	1.2	0.92
Brix value	19.8	21.6	1.1	0.28
Colostrum yield,[1] L/d	3.7	4.4	0.5	0.36
IgA, µg/mL	1,133	1,089	115	0.80
IgM, µg/mL	5,731	5,469	612	0.78
IgG, mg/mL	108	95	18	0.64

[1]First colostrum samples were taken 2.2 ± 1.9 h after calving.

Table 4. Effect of D_3 (n = 20) or HyD (n = 19) supplementation in the close-up diet of the cows on plasma concentrations of fat-soluble vitamins, Ig, and metabolites in newborn calves

	Treatment			
Item	D_3	HyD	SEM	*P*-value
25-OH-D_3, ng/mL	8.91	13.88	0.85	<0.01
25-OH-D_2, ng/mL	1.73	1.76	0.15	0.88
Vitamin A, µg/mL	0.14	0.15	0.01	0.42
Vitamin E, µg/mL	2.13	2.24	0.15	0.64
IgA, µg/mL	358	284	51	0.35
IgM, µg/mL	1,471	1,485	188	0.96
IgG, mg/mL	23.4	22.5	3.2	0.84
Total protein, g/L	62.2	63.7	1.9	0.59
Creatinine, µ*M*	74.5	75.9	2.8	0.74
Total Ca, m*M*	2.21	2.13	0.16	0.76
Inorganic P, m*M*	2.33	2.38	0.06	0.66
Mg, m*M*	0.67	0.65	0.02	0.50

disorder in the entire experiment (Jensen et al., 2026). The discrepancy in sampling location (plasma vs. infected mammary gland) may explain the contradictory findings between our results and those reported by Nelson et al. (2010b), as vitamin D showed localized direct effects on immune cells (Vieira-Neto et al., 2021). In support of our results, Nelson et al. (2010b) found no change in upregulation of 1α-hydroxylase enzyme and vitamin D–binding protein of healthy mammary gland; however, in the *Streptococcus uberis*–infected mammary gland, vitamin D signaling was triggered as part of the innate immune defense against pathogens, enabling localized regulation of immune activity dependent on vitamin D. In contrast to our findings, Poindexter et al. (2020) reported that dietary supplementation of healthy cows with 3 mg of 25-OH-D_3 compared to 1 mg of D_3 reduced the expression of the adhesion protein CD11b on peripheral blood leukocytes. This reduction in CD11b may help regulate or suppress systemic inflammation. It has been well-documented that the mode of action of D_3 or HyD involves the expression of 1α-hydroxylase in immune cells, which regulates the local production of $1,25(OH)_2D_3$ (Waters et al., 2001). In the present study, similar plasma concentrations of $1,25(OH)_2D_3$ in D_3 and HyD groups may explain the lack of response in immune parameters (Jensen et al., 2026). The $1,25(OH)_2D_3$-induced changes in immune cell proliferation are associated with cellular differentiation (Huber et al., 2014). However, in the current study, the cells in peripheral blood of healthy cows may have lacked the necessary stimulation to proliferate (Nonnecke et al., 1993).

Overall, the plasma TNF-α was lower in HyD group compared to D_3 group except for d 20 relative to calving. The TNF-α produced by macrophages originate from monocytes. In line with our findings, Martinez et al. (2018) reported a numerical decrease in monocyte count in cows fed a diet with a DCAD of −130 mEq/kg, containing 3 mg of 25-OH-D_3 compared to those receiving 3 mg of D_3. High concentrations of plasma 25-OH-D_3 in HyD may have directly influenced the vitamin D receptor, stimulating immune cell activity independently of systemic fluctuations in $1,25(OH)_2D_3$ or its cellular synthesis and associated autocrine functions (Lou et al., 2010). Lower plasma TNF-α concentrations in the HyD group compared to the D_3 group could be explained by the down-regulation of T-helper 17 cell expression, which is involved in IL-17 production. Although T-helper 17 cells are essential for promoting immune responses against certain pathogens, they also contribute to inflammatory reactions in tissues (Bettelli et al., 2007; Korn et al., 2007).

Plasma Concentrations of Vitamin A and E in Cows

Jensen et al. (1999) investigated changes in plasma retinol and α-tocopherol concentrations in Holstein–Friesian cows, and, consistent with our findings, they reported a pronounced decline in both vitamins during the first week of lactation. In our study, plasma vitamin A concentration on d 1 relative to calving was comparable to findings reported by Jensen et al. (1999; ~0.23 µg/mL), whereas plasma vitamin E concentration was higher than previously reported (~2.0 µg/mL). Plasma vitamin A concentration decreased during late gestation and reached a nadir on d 1 relative to calving. An optimal range of vitamin A plasma concentration has not been established for cattle but is assumed to be similar to the optimal range in human studies. Plasma vitamin A concentration of 0.2 µg/mL is considered a threshold concentration (WHO, 2009). Similar to vitamin A, plasma vitamin E concentration reached its lowest concentration on d 1 relative to calving. A plasma vitamin E concentration of 2.0 to 3.0 mg/mL is considered a marginal recommended concentration in adult cows. In agreement with our results, Goff and Stabel (1990) demonstrated a 55% and 53% reduction in vitamins A and E, respectively, the first day after calving compared to 14 d prepartum. A study by Goff et al.

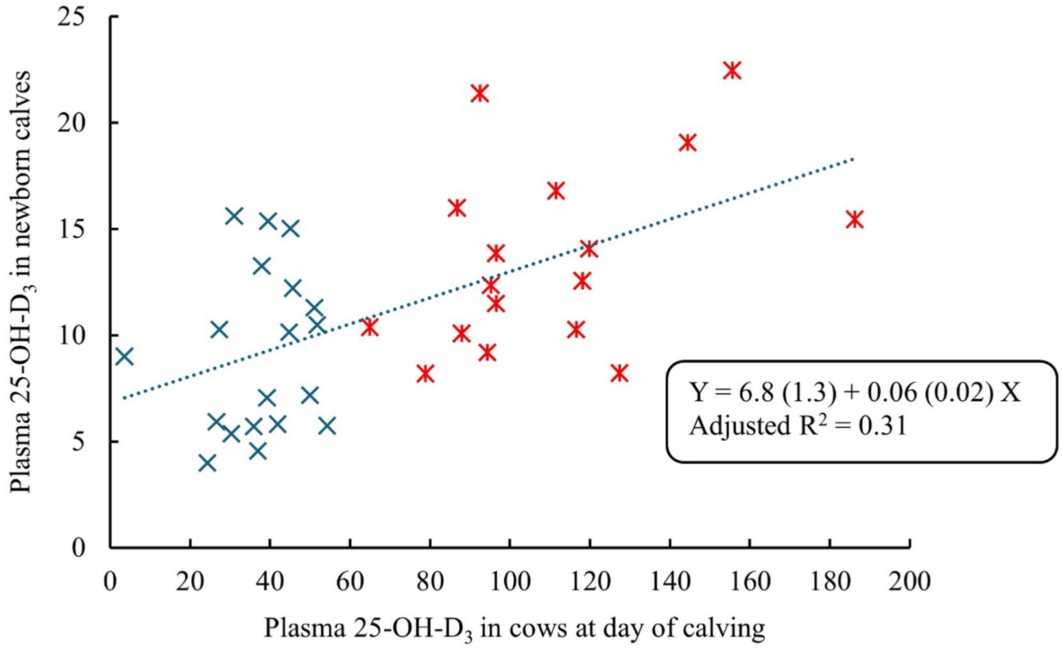

Figure 4. Regression between plasma concentration of 25-OH-D_3 in newborn calves and their dam across D_3 (×) and HyD (ж) groups. Numbers in brackets represent the SE for intercept and slope. Intercept ($P < 0.01$) and slope ($P < 0.01$) were different from zero. There was no significant difference between treatments.

(2002) showed that plasma vitamins A and E were higher in mastectomized cows compared to intact cows at d 0, 5, 10, and 15, which confirms the theory that the decrease in plasma concentrations during late gestation is mainly due to mammary uptake of vitamins A and E for colostrogenesis (Goff and Stabel, 1990; Lashkari et al., 2025). On d 20 relative to calving, plasma vitamins A and E reached a concentration above the threshold for deficiency, which could be due to increased DMI and lower vitamin A and E content in milk compared to colostrum (Lashkari et al., 2025). Jensen et al. (1999) demonstrated the same pattern in plasma vitamin A and E concentrations, and by advancing DIM, the plasma vitamin A and E concentrations increased from the first to the 210th DIM and reached a plateau after 210 DIM.

Plasma Concentrations of Fat-Soluble Vitamins in Newborn Calves

The results of the present study showed that newborn calves had low plasma concentrations of fat-soluble vitamins in their blood. Circulation of vitamins A, D, and E in blood is closely linked to retinol-binding protein (Böhles, 1997), α-tocopherol transport protein (Pazak and Scholz, 1996), and vitamin D–binding protein (Fernando et al., 2020), respectively, and partly linked to lipoproteins (Combs and McClung, 2016). The mammalian fetus is recognized by a low vitamin A, E, and D status due to a low synthesis of retinol-binding protein (Böhles, 1997), α-tocopherol transport protein (Pazak and Scholz, 1996), and vitamin D–binding protein (Bouillon et al., 1977; Fernando et al., 2020). The presence of a substantial placental barrier impeding the transfer of fat-soluble vitamins has been well-documented (Leat, 1967; Pazak and Scholz, 1996). Additionally, plasma of newborn ruminants contains very low concentrations of fat due to limited placental transfer of fat from maternal to fetal circulation (Leat, 1967; Pazak and Scholz, 1996); thus, it may limit the transfer of vitamins A, E, and D via lipoproteins. The low plasma 25-OH-D_2 concentrations in newborn calves may be explained by limited transfer of lipoproteins from maternal to fetal circulation. Hymøller and Jensen (2016) showed that 25-OH-D_2 is exclusively bound to and circulates with a mixture of low- and high-density lipoproteins. The low concentrations of fat-soluble vitamins may reflect the limited tolerance to these vitamins or reduced expression of binding proteins involved in their absorption, transport, and delivery in fetus and newborn calves, or both. Bouillon et al. (1977) found that the blood concentration of vitamin D–binding protein and total 25-OH-D_3 was significantly higher in mothers compared to fetuses, indicating that despite a high maternal-fetal gradient for 25-OH-D_3 transport, fetuses had low vitamin D concentrations. In addition, fetal cord blood had a slightly higher concentration of free 25-OH-D_3 compared to maternal serum, suggesting that the transfer of the carrier protein, rather than the transfer of vitamin D itself, could be responsible for low vitamin D concentrations in neonatal plasma of calves delivered following normal pregnancies (Bouillon et al., 1977; Fernando et al., 2020).

The role of vitamin D in Ca and P metabolism has been well established (Combs and McClung, 2016). However, despite the significant difference in plasma 25-OH-D_3 concentrations, there was no significant difference in plasma Ca and inorganic P concentrations between the D_3 and HyD groups in newborn calves. Different Ca and P regulatory mechanisms in the dam and fetus may explain the low 25-OH-D_3 concentration in the fetus compared to their dam. Goff et al. (1982) suggested that the placenta's ability to indirectly regulate fetal plasma Ca and P concentrations is partly dependent on the maternal 1,25$(OH)_2D_3$ status. Goff et al. (1982) found no correlation between neonatal plasma concentrations of Ca and 1,25$(OH)_2D_3$, indicating that neonatal 1,25$(OH)_2D_3$ may not play a major role in the placental transfer of Ca and P. Supporting the earlier study by Goff et al. (1982), Ashley et al. (2022) reported that only 27% of 1,25$(OH)_2D_3$ from the maternal circulation is taken up by the placenta. Out of this 27%, 22% is retained in the placenta, and 5% is transferred to the fetal blood circulation (Ashley et al., 2022). The majority of 25-OH-D_3 is converted to 1,25$(OH)_2D_3$ by the activity of placental 1α-hydroxylase. Interestingly, 67% of the total 1,25$(OH)_2D_3$ produced in the placenta is retained there, 28% is transferred to the maternal blood circulation, and only 5% is transferred to the fetal blood circulation (Ashley et al., 2022). Thus, such a negligible transfer of both 25-OH-D_3 and 1,25$(OH)_2D_3$ may support the earlier hypothesis proposed by Goff et al. (1982) that maintaining plasma Ca and P concentrations in the fetus depends on a sufficient maternal 1,25$(OH)_2D_3$ status. According to the results of the present study, it seems that the concentration of 25-OH-D_3 has a limited role in regulating plasma Ca and inorganic P during the first 48 h after birth, most likely due to high availability of Ca and P from colostrum, transition milk, and milk. A limited and tightly controlled placental transfer of vitamin D justifies the high concentration of vitamin D (Foley and Otterby, 1978) in colostrum (0.02–0.046) compared to that of mature milk (0.01 µg/g fat) and it highlights the importance of a sufficient supply of vitamin D to newborn calves during the early stage of life.

Even though the optimal concentration for vitamin D status of newborn calves is not known, a concentration of 30 to 50 ng/mL for plasma 25-OH-D_3 concentration is generally recommended as the optimal concentration (NASEM, 2021). The results of the present study showed that all the newborn calves had a lower 25-OH-D_3 than the recommended concentration in adult animals (Nelson et al., 2012, 2016b). Nelson et al. (2016b), in a comprehensive study, demonstrated a high prevalence of low vitamin D concentration in newborn beef calves in the United States regardless of season and geographical location. The present study showed greater plasma 25-OH-D_3 concentration of HyD group compared to D_3 group. Similar to our results, Weiss et al. (2015) reported a higher plasma 25-OH-D_3 concentration in newborn calves from cows fed 6 mg of 25-OH-D_3 compared to cows fed 0.45 mg/d/cow of D_3 during the last 13 d of gestation. Additionally, other studies have shown a correlation between plasma 25-OH-D_3 concentration in newborn calves and their dams (Goff et al., 1982; Weiss et al., 2015; Nelson et al., 2016b), indicating that the serum 25-OH-D_3 status of the dam is reflected in the 25-OH-D_3 status in newborn calves. It is important to note that part of the increased plasma 25-OH-D_3 in calves could be due to a higher uptake of 25-OH-D_3 from colostrum and transition milk in the HyD group during the first 48 h after birth, which was not measured in the current study. Guo et al. (2018) reported that milk 25-OH-D_3 concentrations increased from 869 to 1,001 (ng/kg) in cows receiving a diet containing 0.625 mg/d of 25-OH-D_3 from 14 d precalving to 56 d postcalving compared to cows receiving 0.625 mg of D_3 from 14 d precalving until calving and followed by a combination of 0.625 mg of D_3 and 1.5 mg of 25-OH-D_3 from calving to 56 d postcalving.

Concentrations of Colostrum Ig, Plasma Ig, and Plasma Metabolites in Newborn Calves

The results of the present study showed no effect of treatment on the colostrum and plasma IgA, IgM, and IgG concentrations. Similar to our finding, Poindexter et al. (2023) reported no significant effect of 25-OH-D_3 supplementation on colostrum IgA. However, Martinez et al. (2018) reported a higher colostrum IgG concentration in cows fed a diet with DCAD of −128 mEq/kg of DM containing 3 mg/d of 25-OH-D_3 compared to 3 mg/d of D_3 from 252 d of gestation until calving. Sakem et al. (2013) found that plasma 25-OH-D_3 and Ig concentrations are associated. However, in the present study, despite significantly higher plasma 25-OH-D_3 concentrations in the cows on the day of calving, there was no significant treatment effect on colostrum and plasma concentrations of IgA, IgM, or IgG. In line with our findings, plasma concentrations of Hap, SAA, IFN-γ, and IL-10 before and after calving were unaffected by treatment in the same cows receiving either D_3 or HyD, except for TNF-α. Vitamin D metabolites have been shown to interact with vitamin D receptors expressed on B cells, triggering apoptosis in activated B cells. This action suppresses their proliferation and differentiation, ultimately leading to reduced Ig synthesis (Chen et al., 2007). However, this suppressive effect does not impact inactive B cells or existing plasma cells (Chen et al., 2007). No incidences of disease were recorded among the cows during the close-up period of our study. Therefore, a higher plasma 25-OH-D_3 concentration likely had no effect on Ig secretion, as most B cells in healthy cows were not in an activated state (Chen et al., 2007; Yue et al., 2018).

CONCLUSIONS

Feeding 25-OH-D_3 from 21 d before the expected calving until 21 d after calving induced a lower plasma TNF-α concentration prepartum in the cows. Supplementation of 25-OH-D_3 during the transition period did not impact plasma Hap, SAA, IFN-γ, or IL-10 concentrations in the cows. Additionally, the increased plasma 25-OH-D_3 in cows did not impact IgA, IgM, and IgG concentrations in colostrum. The results demonstrated that HyD supplementation during the close-up period led to greater plasma 25-OH-D_3 concentrations in newborn calves in HyD group compared to D_3 group. However, plasma and colostrum IgA, IgM, and IgG concentrations in newborn calves were not affected, most likely due to a lack of immune response of the cows to dietary 25-OH-D_3 supplementation during the close-up period. Notably, day relative to calving influenced vitamin A and E concentrations in the cows, with considerable reductions in plasma vitamins A and E around calving.

NOTES

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Nonstandard abbreviations used: 25-OH-D_2 = 25-OH-vitamin D_2; 25-OH-D_3 = 25-OH-vitamin D_3; Hap = haptoglobin; HyD = 25-hydroxycholecalciferol; SAA = serum amyloid A.

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